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CUET UG

Environmental Science (Code 307)

Human Beings and Nature

Ecological Thought, Land Stewardship, Social Ecology, Ecofeminism, and Green Politics

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

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Syllabus Alignment & CUET Exam Weightage

Syllabus Component	Core Theoretical Dimensions & Thinkers	Expected MCQs	Marks Weightage
Modern Ecological Thought	Deep Ecology vs. Shallow Ecology (Arne Næss 1973), Biocentric Egalitarianism, Self-Realization, Radical activism (Earth First!, Dave Foreman).	2 - 3 MCQs	10 - 15 Marks
Stewardship & Land Ethic	Aldo Leopold's Land Ethic (**A Sand County Almanac** 1949), Biotic Community expansion, Wendell Berry's agrarian stewardship, Traditional Ecological Knowledge (TEK).	2 MCQs	10 Marks
Social Ecology & Marxism	Murray Bookchin (**The Ecology of Freedom** 1982), Domination of nature as a consequence of social hierarchies, Libertarian municipalism, Barry Commoner's 4 Laws of Ecology.	2 MCQs	10 Marks
Ecofeminism & Grassroots	Françoise d'Eaubonne (1974), Vandana Shiva (**Staying Alive** 1988), Maria Mies, Patriarchal reductionism, Chipko Movement (Gaura Devi, Bahuguna), Navdanya, Seed sovereignty.	2 - 3 MCQs	10 - 15 Marks
Green Politics & Ethics	Four Pillars of Green Politics (Ecological wisdom, Social justice, Grassroots democracy, Non-violence), German Greens (Petra Kelly), Anthropocentrism vs. Ecocentrism, Rights of Nature.	1 - 2 MCQs	5 - 10 Marks

CUET UG UNIT 1 STRATEGIC EXAM POINTERS

- **1. The Næss Distinction:** Always distinguish Arne Næss's "Shallow Ecology" (reformist, technocratic, anti-pollution for human affluence) from "Deep Ecology" (biocentric egalitarianism, inherent value of all organisms independent of human use).
- **2. Leopold's Golden Rule:** Memorize Leopold's exact normative test: **"A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong otherwise."**
- **3. Root Cause Attribution:** Social ecology (Bookchin) traces environmental destruction to **human social hierarchy and capitalism**, NOT human nature or overpopulation per se.
- **4. Ecofeminist Authorship:** Associate **Staying Alive** and seed sovereignty with **Vandana Shiva**; the term "écoféminisme" was coined by **Françoise d'Eaubonne** in 1974.

1. Modern Schools of Ecological Thought: Deep vs. Shallow Ecology

1.1 The Genesis of Deep Ecology (Arne Næss)

The modern philosophy of environmentalism experienced an ontological shift in 1973 when Norwegian philosopher **Arne Næss** published his seminal paper, *"The Shallow and the Deep, Long-Range Ecology Movement: A Summary"* in the journal **Inquiry**. Næss classified environmental thought into two fundamentally divergent movements:

1. **The Shallow (Reformist) Ecology Movement:**
   - **Core Premise:** Anthropocentric (human-centered). It views nature strictly as a storehouse of resources and a waste sink for human industrial civilization.
   - **Primary Objectives:** Fighting pollution, mitigating resource depletion, and stabilizing climatic disruptions solely to safeguard the health, economic affluence, and living standards of developed societies.
   - **Methods:** Technological optimization, "end-of-pipe" scrubbers, carbon offset accounting, market-based environmental regulations, and resource efficiency within existing capitalist structures.
   - **Critique:** Superficial; treats symptoms without addressing systemic industrial-consumerist roots.
2. **The Deep Ecology Movement:**
   - **Core Premise:** Ecocentric and Biocentric Egalitarian. It rejects the Cartesian dualism that separates humans from the natural ecosystem.
   - **Ontological Axiom:** Non-human life forms—including animals, plants, rivers, mountains, and microbiological biota—possess **intrinsic value (inherent worth)** completely independent of their economic, nutritional, or aesthetic utility to human beings.
   - **Philosophical Foundation:** Synthesizes Spinozan metaphysics, Mahayana Buddhism, Gandhi's non-violence (**Ahimsa**), and holistic systems ecology.

1.2 Biocentric Egalitarianism and Self-Realization

Næss established two primary ultimate norms (**intuitions**) that underpin deep ecological consciousness:

- **Biocentric Egalitarianism in Principle:** The recognition that all living organisms on Earth are interrelated members of an unbroken biosphere and have an equal moral right to live, blossom, and reach their individual forms of unfolding. Næss added the qualification "in principle" to acknowledge that humans inevitably must consume other living organisms to survive, but such consumption must be restricted to vital survival needs rather than frivolous consumer luxury.
- **Self-Realization (Capital 'S'):** Contrast with narrow egoistic self-gratification (small 's'). Self-Realization in deep ecology involves an expansive identification of the human self with the totality of the living ecosphere. When an individual achieves ecological Self-Realization, harming a wetland or clear-cutting an old-growth forest is experienced as an assault upon one's own bodily organism.

1.3 The 8-Point Deep Ecology Platform (Næss and Sessions, 1984)

In 1984, during a retreat in Death Valley, California, Arne Næss and American philosopher **George Sessions** formulated the definitive **8-Point Deep Ecology Platform**:

1. The well-being and flourishing of human and non-human life on Earth have value in themselves (inherent value). These values are independent of the usefulness of the non-human world for human purposes.
2. Richness and diversity of life forms contribute to the realization of these values and are also values in themselves.
3. Humans have no right to reduce this richness and diversity except to satisfy **vital needs**.
4. The flourishing of human life and cultures is compatible with a substantial decrease of the human population. The flourishing of non-human life requires such a decrease.
5. Present human interference with the non-human world is excessive, and the situation is rapidly worsening.
6. Policies must therefore be changed. These policies affect basic economic, technological, and ideological structures. The resulting state of affairs will be deeply different from the present.
7. The ideological change is mainly that of appreciating life quality (dwelling in situations of inherent value) rather than adhering to an increasingly higher standard of living. There will be a profound awareness of the difference between big and great.
8. Those who subscribe to the foregoing points have an obligation directly or indirectly to participate in the attempt to implement the necessary changes.

1.4 Radical Activism: Earth First! and Direct Action

Deep ecological philosophy directly inspired the radical ecological movement **Earth First!**, founded in 1980 by **Dave Foreman**, Mike Roselle, and Howie Wolke in the southwestern United States:

- **Motto: "No Compromise in Defense of Mother Earth!"**
- **Tactics:** Pioneered non-violent direct action, civil disobedience, and **monkeywrenching** (strategic ecological sabotage such as tree-spiking, billboard defacement, and disabling heavy logging bulldozers) to protect wilderness corridors and roadless areas.
- **Influence of Edward Abbey:** Abbey's 1975 novel, **The Monkey Wrench Gang**, provided the imaginative template for biocentric resistance against dam construction (Glen Canyon Dam) and strip mining.

Modern Schools of Ecological Thought: Philosophical Comparative Matrix

Ontological Premises, Key Theorists, Foundational Texts, Axiological Principles & Critiques

CUET UG · EVS UNIT 1 · FIG 1

Ecological School	Key Theorists & Origins	Foundational Literature	Core Philosophical Tenet	CUET High-Yield Focus
Deep Ecology	Arne Næss (1973), George Sessions, Gary Snyder	*The Shallow and the Deep, Long-Range Ecology Movement*	Biocentric Egalitarianism; intrinsic value of all living entities independent of human utility; self-realization beyond ego.	Rejection of anthropocentrism; 8-point platform; Earth First! activism.
Shallow / Reform Ecology	Mainstream technocrats, Resource managers	*Brundtland Report (Our Common Future, 1987)*	Anthropocentric environmentalism; fighting pollution & resource depletion primarily to safeguard human affluence and health.	End-of-pipe tech fixes; market instruments; contrasts sharply with Deep Ecology.
Social Ecology	Murray Bookchin (1960s-1980s)	*The Ecology of Freedom (1982), Post-Scarcity Anarchism*	Ecological crises stem directly from human-on-human hierarchical domination, social inequality, and capitalist exploitation.	Rejects biocentrism as anti-human; advocates libertarian municipalism.
Ecofeminism	Françoise d'Eaubonne (1974), Vandana Shiva, Maria Mies	*Staying Alive (Shiva, 1988), Ecofeminism (Mies & Shiva)*	Interconnected domination of nature and women under patriarchal-capitalist systems; nurturance vs violent reductionism.	Chipko movement connection; seed sovereignty (Navdanya); critique of Green Revolution.
Land Ethic / Stewardship	Aldo Leopold (1949), Wendell Berry	*A Sand County Almanac, The Unsettling of America*	"A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong otherwise."	Enlarges boundary of community to include soils, waters, plants; ethical stewardship.
Green Politics / Ecosocialism	Barry Commoner, Petra Kelly (German Greens)	*The Closing Circle (1971), Thinking Green!*	Four Pillars: Ecological wisdom, social justice, grassroots democracy, non-violence. First Law: "Everything is connected to everything else."	Commoner's 4 Laws of Ecology; institutionalizing green legislative mandates.

OURMOCKS CUET UG ENVIRONMENTAL SCIENCE SERIES CODE 307 · UNIT 1: HUMAN BEINGS & NATURE PAGE 5 CONCEPT MATRIX

Figure 1: Modern Schools of Ecological Thought: Philosophical Comparative Matrix

2. Land Ethic and Stewardship: From Aldo Leopold to Wendell Berry

2.1 Aldo Leopold and the Evolution of Environmental Ethics

Forester, wildlife biologist, and ecologist **Aldo Leopold** (1887–1948) provided the conceptual foundation for modern environmental ethics in his 1949 masterpiece, **A Sand County Almanac**:

- **Historical Sequence of Ethics:** Leopold argued that human ethical frameworks evolved through three distinct evolutionary stages:
  1. **Stage 1 (Inter-Individual Ethics):** Deals with relations between individuals (e.g., the Ten Commandments, Mosaic Decalogue).
  2. **Stage 2 (Individual-Society Ethics):** Deals with relations between the individual and the civic community (e.g., the Golden Rule, democratic social contracts).
  3. **Stage 3 (The Land Ethic - Missing Horizon):** Deals with human relations to the land, soils, waters, fauna, and flora. Historically, land was treated strictly as private property with economic privileges but zero moral obligations (analogous to the slave-girls of Odysseus, whom he hanged without moral inquiry because they were chattel).
- **The Biotic Community:** The Land Ethic fundamentally **"enlarges the boundaries of the community to include soils, waters, plants, and animals, or collectively: the land."**
- **Role Transformation:** It transforms the role of *Homo sapiens* from conqueror of the land-community to plain member and citizen of it. It implies respect for fellow members and respect for the community as such.
- **The Golden Normative Test:**

> **ALDO LEOPOLD'S LAND ETHIC AXIOM (1949)**
>
> **"A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong otherwise."**

- **Thinking Like a Mountain:** In the chapter titled *Thinking Like a Mountain*, Leopold recounted killing an old mother wolf in New Mexico. Watching the "fierce green fire" die in her eyes, he realized that eliminating apex predators destroys the mountain's ecological equilibrium: overpopulated deer herds starve the mountainside, strip the brush, and trigger catastrophic soil erosion.

2.2 Wendell Berry and Agrarian Stewardship

Agrarian essayist, poet, and farmer **Wendell Berry** extended Leopold's land ethic into human food production, community economics, and agricultural stewardship:

- **Key Texts: The Unsettling of America: Culture & Agriculture** (1977), **Home Economics** (1987).
- **Critique of Industrial Agribusiness:** Berry fiercely attacked modern fossil-fuel-intensive agribusiness (embodied by Earl Butz's "get big or get out" policy) for reducing complex biological living soils into industrial extraction factories.
- **Stewardship Principles:**
  - True stewardship requires an enduring domestic affection for specific geography (**topophilia**), local community memory, and intergenerational care.
  - Eating is an agricultural and ecological act: how food is produced, transported, and consumed reflects human ethical alignment with biological ecosystems.

- The crisis of nature is a crisis of character: displaced farmers are severed from their biological feedback loops, turning land into an alienated monoculture.

2.3 Traditional Ecological Knowledge (TEK) and Indigenous Stewardship

Long before formal environmental philosophy emerged, indigenous and traditional forest-dwelling communities practiced sophisticated systems of natural resource stewardship:

- **Sacred Groves (Orans in Rajasthan, Devarakadu in Karnataka, Sarnas in Jharkhand):** Patches of climax forest dedicated to local deities, where all logging, hunting, and grazing are strictly prohibited by customary taboos, serving as vital micro-refugia for endemic biodiversity and watershed preservation.
- **Bishnoi Community of Rajasthan:** Founded in the 15th century by **Guru Jambheshwar (Jamboji)**, who established 29 environmental and moral commandments. In 1730 AD, **Amrita Devi Bishnoi** and 363 Bishnoi villagers sacrificed their lives clinging to sacred **Khejri trees (Prosopis cineraria)** at Khejarli village to protect them from the axes of the Maharaja of Jodhpur's soldiers. This historical sacrifice became the direct forerunner to the Chipko movement.

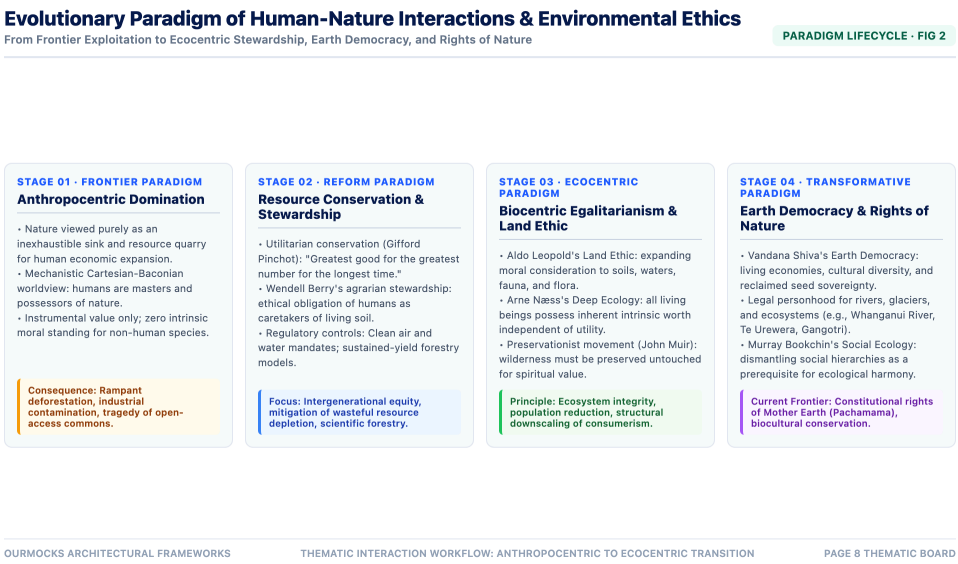


Figure 2: Evolutionary Paradigm of Human-Nature Interactions & Environmental Ethics

3. Social Ecology and Marxist Environmentalism

3.1 Murray Bookchin's Social Ecology

American political philosopher **Murray Bookchin** (1921–2006) founded the school of **Social Ecology**, publishing cornerstone works including **Our Synthetic Environment** (1962), **Post-Scarcity Anarchism** (1971), and **The Ecology of Freedom: The Emergence and Dissolution of Hierarchy** (1982):

- **The Central Thesis:** All contemporary ecological crises (deforestation, toxic pollution, climatic collapse, biodiversity loss) do not stem from human biology or innate human destructiveness, but are the direct systemic product of **human-on-human hierarchical domination**.
- **The Mechanism of Domination:** The psychological and social notion that humans must dominate the natural world arose historically from the prior institutionalization of the domination of human by human:
  - Elders dominating youth (gerontocracy);

- Men dominating women (patriarchy);
- Nobles dominating peasants (feudalism);
- Capitalists dominating wage laborers (industrial capitalism).

- **Critique of Deep Ecology:** Bookchin launched a scathing critique of deep ecology, accusing it of reactionary **misanthropy** and biological reductionism. He argued that blaming "humanity as a whole" for ecological destruction ignores the reality that indigenous peoples, impoverished peasants, and urban working classes do not direct or profit from multinational oil corporations or timber syndicates.
- **Libertarian Municipalism (Communalism):** Bookchin proposed replacing the centralized nation-state and capitalist market economy with directly democratic, confederated citizen assemblies practicing eco-technologies and decentralized municipal self-governance.

3.2 Barry Commoner and the Critique of Production Technology

Cellular biologist and environmentalist **Barry Commoner** (1917–2012), dubbed the "Paul Revere of Ecology," synthesized socialist economics with ecological systems science:

- **The Closing Circle (1971):** Commoner demonstrated through rigorous empirical analysis that post-World War II environmental degradation in the United States was NOT primarily caused by population growth or per capita affluence, but by the displacement of ecologically benign natural technologies by synthetic, non-biodegradable petrochemical technologies (e.g., replacing cotton with polyester, soap with synthetic detergents, glass with plastics, and organic manures with inorganic nitrogen fertilizers).
- **Commoner's Four Laws of Ecology:**
  1. **Everything is Connected to Everything Else:** The biosphere is an intricate, non-linear web of interdependent geochemical and biological cycles. Disrupting one node produces cascading unintended consequences.
  2. **Everything Must Go Somewhere:** There is no "waste" or "away" in nature. Every synthetic chemical, heavy metal, or plastic emitted into the environment persists and circulates through trophic food chains.
  3. **Nature Knows Best:** Natural organic chemical compounds have evolved over 3.5 billion years of trial and error. Synthetic anthropogenic molecules foreign to natural biochemical pathways (e.g., DDT, PCBs, CFCs) are almost universally toxic, persistent, or disruptive to metabolic enzymes.
  4. **There is No Such Thing as a Free Lunch:** Every technological and economic gain extracted from nature incurs an ecological cost that must ultimately be paid, whether through soil depletion, disease, or planetary instability.

4. Ecofeminism: Intersecting Gender, Patriarchy, and Ecological Defense

4.1 Emergence and Core Philosophies of Ecofeminism

The term **écoféminisme** was coined in 1974 by French feminist author **Françoise d'Eaubonne** in her book *Le Féminisme ou la Mort* (Feminism or Death). Ecofeminism investigates the profound structural, historical, and ideological connections between the patriarchal oppression of women and the capitalist-industrial exploitation of nature:

- **Carolyn Merchant's The Death of Nature (1980):** Merchant traced how the 16th and 17th-century Scientific Revolution (Francis Bacon, René Descartes) dismantled the ancient organic cosmology of the Earth as a living, nurturing Mother (**Anima Mundi**) and replaced it

with a mechanistic, reductionist worldview. Baconian science explicitly framed nature as a wild female force to be "bound into service," interrogated on the rack, and enslaved for masculine technological mastery.

- **Dualistic Hierarchies:** Ecofeminist philosopher Val Plumwood demonstrated how Western philosophical thought constructed oppressive dualisms:
  - Culture / Nature
  - Reason / Emotion
  - Male / Female
  - Mind / Body
  - Human / Non-Human

  In every binary, the former is coded masculine, superior, and rational, while the latter is coded feminine, inferior, passive, and legitimate to conquer.

4.2 Vandana Shiva, Maria Mies, and the Indian Grassroots Context

Physicist and environmental activist **Dr. Vandana Shiva**, in seminal works such as **Staying Alive: Women, Ecology and Development** (1988) and **Ecofeminism** (co-authored with German sociologist **Maria Mies**, 1993), exposed the neo-colonial and reductionist character of industrial development:

- **Critique of the Green Revolution:** Shiva argued that the Green Revolution introduced a reductionist, masculine, capital-intensive technology based on chemical poisons (pesticides), synthetic fertilizers, and water-guzzling monocultures. This destroyed biodiversity, depleted ancient aquifers, contaminated soils, and displaced peasant women, who were historically the custodians of seed diversity, post-harvest processing, and agroecological wisdom.
- **Prakriti and the Feminine Principle:** Drawing upon traditional Indian cosmology, Shiva posited the concept of **Prakriti**—nature as an active, creative, living, self-organizing female energy. Development projects that ignore the feminine principle become projects of **maldevelopment**, generating poverty, ecological desertification, and female marginalization.
- **Navdanya and Seed Sovereignty (Bija Swaraj):** Founded in 1987 by Vandana Shiva, Navdanya created a network of community seed banks across India, conserving over 5,000 indigenous rice and crop varieties, championing chemical-free organic farming, and legally defeating corporate patents on indigenous genetic resources (such as W.R. Grace's attempt to patent Neem and RiceTec's attempt to patent Basmati rice).

4.3 The Chipko Movement: Vanguard of Eco-Feminist Resistance

The **Chipko Movement** (from Hindi *chipko*, meaning "to embrace" or "to hug") emerged in the Garhwal Himalayas of Uttarakhand in the early 1970s:

- **The Catalytic Event (Reni Village, March 1974):** When state forest contractors arrived to log 2,455 trees in the Reni forest while village men were away in Chamoli for compensation hearings, **Gaura Devi**, a 50-year-old illiterate peasant woman, mobilized 27 village women. They confronted armed loggers, clung physically to the tree trunks, and declared the forest their maternal home: **"This forest is our mother's home; we will not let you axe it."** The loggers were forced to retreat.
- **Leadership and Slogans:** While grassroots rural women were the movement's operational backbone, legendary Gandhian activists **Sunderlal Bahuguna** and **Chandi Prasad Bhatt** (founder of Dasholi Gram Swarajya Mandal - DGSM) provided leadership and philosophical framing.
- **The Iconic Chipko Slogan (coined by village women):**

> **THE HISTORIC CHIPKO FOREST SLOGAN**

> *What do the forests bear? Soil, water, and pure air! Soil, water, and pure air, are the basis of sustaining life!*

- **Direct Policy Impact:** In 1980, following Sunderlal Bahuguna's 5,000-kilometer trans-Himalayan padayatra and meeting with Prime Minister Indira Gandhi, the Indian government enacted a 15-year total ban on commercial green tree felling in the Himalayan forests above 1,000 meters altitude.

5. Green Politics, Environmental Ethics, and Earth Jurisprudence

5.1 The Four Pillars of Green Politics

Green Politics emerged as a distinct parliamentary political ideology in Western Europe, crystalized by the formation of **Die Grünen (The German Green Party)** in 1980 under the pioneering leadership of **Petra Kelly**, Heinrich Böll, and Joseph Beuys. Green Parties worldwide operate upon the foundational **Four Pillars of Green Politics**:

1. **Ecological Wisdom:** Operating human socioeconomic systems strictly within planetary boundaries, respecting natural biogeochemical carrying capacities, conserving biodiversity, and abandoning perpetual growth paradigms.
2. **Social Justice:** Eradication of poverty, equitable global resource distribution, dismantling structural oppression, and securing intergenerational equity.
3. **Grassroots Democracy:** Decentralizing decision-making power down to local community assemblies, participatory budgeting, and transparent governance.
4. **Non-Violence (Peace):** Demilitarization, disarmament, non-violent civil defense, and resolving geopolitical resource conflicts through diplomatic cooperation.

5.2 Ethical Spectrum: Anthropocentrism vs. Ecocentrism

Ethical Paradigm	Ontological Center of Value	Operational Rule in Environmental Decision-Making
Anthropocentrism	Human beings exclusively	Nature has purely instrumental value (economic, medical, recreational). Non-human degradation is only objectionable if it harms human welfare or future human generations.
Zoocentrism / Sentientism	Sentient animals (Peter Singer, Tom Regan)	Moral standing extends to all organisms capable of experiencing pleasure and suffering (sentience). Animal welfare, opposition to factory farming and vivisection.
Biocentrism	All living individual organisms (Albert Schweitzer, Paul Taylor)	Every individual living organism possesses a "teleological center of life" and inherent worth. Reverence for all biological life regardless of cognitive sentience.
Ecocentrism / Holism	Total ecosystems, biotic communities (Aldo Leopold, Arne Næss)	Value inheres in ecological wholes: species populations, watersheds, rivers, soils, and the biosphere. Individual welfare is subordinated to ecosystem stability and integrity.

5.3 Earth Jurisprudence and the Legal Rights of Nature

Traditional Western legal systems classified all non-human nature as inert **property** without legal standing. However, 21st-century environmental jurisprudence is pioneering the legal recognition of **Rights of Nature (Wild Law)**:

- **Christopher Stone's Precedent (1972):** In his groundbreaking law review essay, *"Should Trees Have Standing? Toward Legal Rights for Natural Objects"*, Stone argued that just as corporations, trusts, and municipalities (inanimate legal fictions) have legal personhood, ecosystems should have legally enforceable rights, represented by human guardians in court.
- **Ecuador's Constitution (2008):** Ecuador became the first nation in world history to constitutionally guarantee the **Rights of Pachamama (Mother Earth)**: the right to exist, persist, regenerate its vital cycles, and receive comprehensive restoration.
- **Whanganui River (New Zealand, 2017):** The New Zealand Parliament passed the **Te Awa Tupua Act**, officially recognizing the Whanganui River as a legal person and an indivisible living whole, represented jointly by the Māori iwi and the Crown.
- **Indian Judicial Landmark (Uttarakhand High Court, 2017):** In **Mohd. Salim v. State of Uttarakhand**, the High Court declared the Rivers Ganga and Yamuna, along with their tributaries and glaciers, as legal living entities possessing the rights of a juridical person, appointing court-mandated guardians for their protection (though later stayed by the Supreme Court on administrative liability concerns).

6. CUET High-Yield Invariant Master Table

Theorist / Figure	Year / Movement	Seminal Text / Publication	Core Invariant Fact / Definition
Arne Næss	1973	The Shallow and the Deep Ecology Movement	Coined "Deep Ecology"; Biocentric Egalitarianism; Self-Realization; Intrinsic value.
Aldo Leopold	1949	A Sand County Almanac	Land Ethic; "Thinking Like a Mountain"; Biotic community integrity, stability, beauty.
Murray Bookchin	1982	The Ecology of Freedom	Founded Social Ecology; Ecological destruction rooted in human social hierarchy.
Barry Commoner	1971	The Closing Circle	Four Laws of Ecology; Production technology critique; Ecosocialism.
Françoise d'Eaubonne	1974	Le Féminisme ou la Mort	Coined the term "Ecofeminism" (écoféminisme); Intersecting oppression.
Vandana Shiva	1988	Staying Alive: Women, Ecology & Dev.	Prakriti / Feminine Principle; Critique of Green Revolution; Seed sovereignty.
Rachel Carson	1962	Silent Spring	Catalyst of modern environmental movement; Bioaccumulation of synthetic DDT pesticide.

Theorist / Figure	Year / Movement	Seminal Text / Publication	Core Invariant Fact / Definition
Amrita Devi Bishnoi	1730 AD	Bishnoi Khejri Movement (Khejarli)	Sacrificed life with 363 villagers defending sacred Khejri trees; 29 principles.
Gaura Devi	1974	Chipko Movement (Reni Village)	Mobilized village women to physically embrace trees to halt commercial felling.
Sunderlal Bahuguna	1970s-80s	Chipko Movement & Anti-Tehri Dam	Gandhian environmentalist; Coined "Ecology is permanent economy"; Himalayan padayatra.
Petra Kelly	1980	Die Grünen (German Green Party)	Co-founder of Green Politics; Four Pillars: Ecology, Social Justice, Democracy, Peace.
Garrett Hardin	1968	The Tragedy of the Commons	Unregulated open-access common pool resources face inexorable overexploitation.

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Human Beings and Nature: Synthesis Mindmap

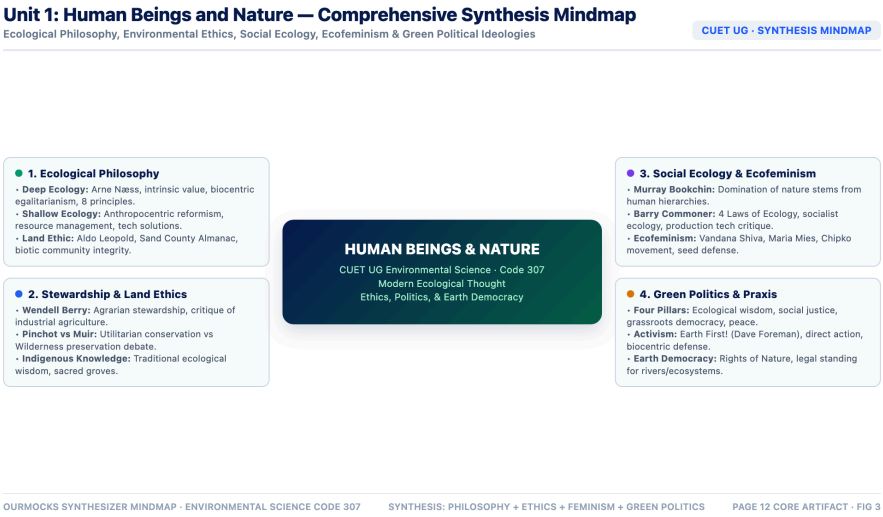


Figure 3: Unit 1: Human Beings and Nature — Comprehensive Synthesis Mindmap

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Human Beings & Nature: Quick Reference Cheatsheet

ESSENTIAL THEORETICAL & HISTORICAL INVARIANTS

▸ **Deep Ecology vs. Shallow Ecology (Arne Næss 1973):** Deep = Ecocentric, intrinsic value of all living entities, population reduction; Shallow = Anthropocentric, technological fixes, fighting pollution for human affluence. ▸ **Aldo Leopold's Land Ethic (1949):** "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong otherwise." Enlarges boundary of community to include soils, waters, plants, and animals. ▸ **Social Ecology (Murray Bookchin 1982):** Ecological crises originate directly from human-on-human hierarchical domination and capitalism. Rejects deep ecology's misanthropy. ▸ **Commoner's 4 Laws of Ecology (1971):** 1. Everything is connected to everything else. 2. Everything must go somewhere. 3. Nature knows best. 4. There is no such thing as a free lunch. ▸ **Ecofeminism (Françoise d'Eaubonne 1974):** Structural link between the subjugation of women and exploitation of nature. Vandana Shiva: **Prakriti**, seed sovereignty (**Bija Swaraj**), Navdanya. ▸ **Four Pillars of Green Politics (Petra Kelly 1980):** Ecological Wisdom, Social Justice, Grassroots Democracy, Nonviolence. ▸ **Historic Resistance:** Bishnoi Sacrifice (1730 AD, Amrita Devi, 363 martyrs, Khejri trees); Chipko Movement (1974, Reni village, Gaura Devi, Sunderlal Bahuguna).

THE SEVEN FATAL CUET TRAPS IN HUMAN BEINGS AND NATURE

▸ **FATAL TRAP 1 (Shallow vs. Deep Ecology Confusion):** Examiners will describe an anti-pollution sewage treatment plant or catalytic converter program and ask for the philosophy. Do NOT mark Deep Ecology! Technocratic fixes to protect human health are the hallmark of **Shallow Ecology**! ▸ **FATAL TRAP 2 (Origin of Social Ecology):** Do NOT attribute Social Ecology to Karl Marx or Barry Commoner. The formal founder of Social Ecology is **Murray Bookchin** (**The Ecology of Freedom** 1982)! ▸ **FATAL TRAP 3 (Coined Term vs. Popularizer):** The term "Ecofeminism" was NOT coined by Vandana Shiva; it was coined by French author **Françoise d'Eaubonne** in 1974 (**Le Féminisme ou la Mort**)! ▸ **FATAL TRAP 4 (Aldo Leopold's Book Title):** Do not confuse Leopold's **A Sand County Almanac** (1949) with Rachel Carson's **Silent Spring** (1962) or Barry Commoner's **The Closing Circle** (1971)! ▸ **FATAL TRAP 5 (Chipko Leadership Nuance):** While Sunderlal Bahuguna and Chandi Prasad Bhatt were the public leaders, the physical defiance at Reni village in March 1974 was led by **Gaura Devi** and 27 rural women! ▸ **FATAL TRAP 6 (Anthropocentrism vs. Ecocentrism):** Utilitarian conservation (Gifford Pinchot: "greatest good for greatest number") is **anthropocentric**, NOT ecocentric. Preservation (John Muir) and Land Ethic (Leopold) are **ecocentric**! ▸ **FATAL TRAP 7 (Deep Ecology on Human Population):** Deep Ecology explicitly advocates for a substantial, voluntary **decrease in the human population** over time to allow non-human life to flourish. Candidates often mistakenly mark this as anti-environmental!

CUET UG Practice Questions: Human Beings and Nature

Q1. According to Norwegian philosopher Arne Næss, what is the fundamental defining principle of the "Deep Ecology" movement?

(A) The intrinsic value of all living entities, independent of their utility to humans
(B) The promotion of technological scrubbers to combat urban air pollution
(C) The maximization of economic yield through sustainable forest logging
(D) The preservation of scenic landscapes primarily for human ecotourism

Sol. Q1. Arne Næss established Deep Ecology in 1973 based on biocentric egalitarianism—the principle that all non-human life has inherent value independent of human economic utility. **Answer: (A)**

Q2. In **A Sand County Almanac** (1949), how did Aldo Leopold formulate his foundational Land Ethic criterion?

(A) A policy is ethical when it maximizes the gross economic product of rural regions
(B) A thing is right when it preserves the integrity, stability, and beauty of the biotic community
(C) Natural resources should be managed strictly for the greatest good of human majorities
(D) Wilderness areas must be fenced off completely to prevent indigenous forest access

Sol. Q2. Leopold's Land Ethic axiom states: "A thing is right when it tends to preserve the integrity, stability, and beauty of the biotic community. It is wrong otherwise." It expands moral standing to soils, waters, plants, and animals. **Answer: (B)**

Q3. Murray Bookchin's school of "Social Ecology" argues that environmental crises are fundamentally rooted in which structural factor?

(A) The biological genetic selfishness inherent in the nature of **Homo sapiens**
(B) The inevitable evolutionary overshoot of carrying capacity across primate species
(C) The institutionalization of human-on-human hierarchical domination and capitalism
(D) The failure of modern agrarian cultures to mandate compulsory vegetarianism

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Sol. Q3. Social Ecology posits that the psychological and institutional impulse to dominate nature arose directly from prior systems of human-on-human domination (gerontocracy, patriarchy, and capitalism). **Answer: (C)**

Q4. In **The Closing Circle** (1971), Barry Commoner formulated Four Laws of Ecology. Which of the following is NOT one of Commoner's laws?

(A) Everything is connected to everything else
(B) Everything must go somewhere
(C) Nature knows best
(D) Economic capital takes precedence over biological cycles

Sol. Q4. Commoner's Four Laws are: 1. Everything is connected to everything else; 2. Everything must go somewhere; 3. Nature knows best; and 4. There is no such thing as a free lunch. Option D directly contradicts ecological principles. **Answer: (D)**

Q5. Who originally coined the term "ecofeminism" (**écoféminisme**) in her 1974 book **Le Féminisme ou la Mort**?

(A) Françoise d'Eaubonne
(B) Vandana Shiva
(C) Carolyn Merchant
(D) Maria Mies

Sol. Q5. The term "écoféminisme" was coined in 1974 by French feminist author Françoise d'Eaubonne. Vandana Shiva, Carolyn Merchant, and Maria Mies subsequently developed the philosophical field. **Answer: (A)**

Q6. In the historic Chipko resistance at Reni village in March 1974, which grassroots female leader mobilized 27 women to physically cling to tree trunks?

(A) Medha Patkar (B) Gaura Devi
(C) Amrita Devi Bishnoi (D) Sunita Narain

Sol. Q6. Gaura Devi, an illiterate 50-year-old Bhotiya woman, led the confrontation at Reni village on March 25, 1974, saving 2,455 trees. Amrita Devi led the 1730 Bishnoi sacrifice, while Medha Patkar led the Narmada movement. **Answer: (B)**

Q7. The historic sacrifice of 363 Bishnoi villagers in 1730 AD to protect sacred trees from the Maharaja of Jodhpur occurred at Khejarli to defend which tree species?

(A) Neem (**Azadirachta indica**) (B) Banyan (**Ficus benghalensis**)
(C) Khejri (**Prosopis cineraria**) (D) Sal (**Shorea robusta**)

Sol. Q7. The Bishnois of Rajasthan, led by Amrita Devi Bishnoi, sacrificed their lives in 1730 AD hugging sacred Khejri (**Prosopis cineraria**) trees, vital for desert soil stabilization and fodder. **Answer: (C)**

Q8. Which international political ideology, pioneered by Petra Kelly and the German Greens in 1980, is formally built upon the "Four Pillars"?

(A) Neoliberal Conservatism (B) Democratic Socialism
(C) Classical Liberalism (D) Green Politics

Sol. Q8. Green Politics (**Die Grünen**) was established on the Four Pillars: Ecological Wisdom, Social Justice, Grassroots Democracy, and Non-violence, providing a framework for Green parties globally. **Answer: (D)**

Q9. In environmental philosophy, what is the critical distinction between "Biocentrism" and "Anthropocentrism"?

(A) Biocentrism grants inherent moral worth to all living organisms; Anthropocentrism centers exclusively on human interests
(B) Biocentrism values only livestock, while Anthropocentrism values only forest wildlife
(C) Biocentrism claims technology solves all shortages; Anthropocentrism advocates hunter-gatherer life
(D) Biocentrism prioritizes minerals, while Anthropocentrism focuses entirely on marine flora

Sol. Q9. Anthropocentrism regards humans as the sole holders of intrinsic moral standing, viewing nature instrumentally. Biocentrism extends moral worth to all individual living organisms. **Answer: (A)**

Q10. Which groundbreaking 1980 text by Carolyn Merchant analyzed how the Scientific Revolution replaced an organic cosmology with a mechanistic worldview?

(A) **Staying Alive** (B) **The Death of Nature**
(C) **Silent Spring** (D) **The Ecology of Freedom**

Sol. Q10. Carolyn Merchant's **The Death of Nature** (1980) analyzed how Cartesian-Baconian mechanistic science legitimized the technological subjugation and exploitation of both nature and women. **Answer: (B)**

Q11. Dr. Vandana Shiva's organization **Navdanya**, founded in 1987, is internationally renowned for advancing which major ecological objective?

(A) Promoting synthetic chemical pesticides and hybrid monoculture seed varieties
(B) Advocating for the privatization of municipal water delivery networks
(C) Conserving indigenous seed biodiversity and defending seed sovereignty (**Bija Swaraj**)
(D) Constructing mega-hydroelectric dams in fragile Himalayan tectonic valleys

Sol. Q11. Navdanya was founded by Vandana Shiva to promote biodiversity conservation, organic farming, and seed sovereignty (**Bija Swaraj**), establishing community seed banks and fighting corporate seed patents. **Answer: (C)**

Q12. What landmark legal doctrine was enacted by New Zealand in 2017 through the **Te Awa Tupua Act**, marking an epochal transition in Earth Jurisprudence?

(A) Permitting unrestricted commercial gravel mining along river basins
(B) Abolishing all tribal fishing rights across North Island freshwater lakes
(C) Designating the entire national coastline as industrial shipping zones
(D) Granting legal personhood and the rights of a living entity to the Whanganui River

Sol. Q12. The Te Awa Tupua Act 2017 recognized the Whanganui River as an indivisible living whole with legal personhood, represented jointly by the Māori iwi and the Crown. **Answer: (D)**

Q13. In the deep ecology platform articulated by Arne Næss and George Sessions in 1984, what demographic principle is explicitly affirmed?

(A) A substantial, humane decrease in the global human population
(B) A mandatory doubling of the human workforce in heavy industrial sectors
(C) The relocation of all human populations into mega-density offshore artificial islands
(D) The unrestricted expansion of agricultural acreage into old-growth equatorial rainforests

Sol. Q13. Point 4 of the Deep Ecology Platform affirms: "The flourishing of human life and cultures is compatible with a substantial decrease of the human population. The flourishing of non-human life requires such a decrease." **Answer: (A)**

Q14. What famous ecological metaphor did Aldo Leopold employ in **A Sand County Almanac** to illustrate the consequences of exterminating apex predators?

(A) "The Tragedy of the Open Commons"
(B) "Thinking Like a Mountain"
(C) "The Invisible Hand of the Biosphere"
(D) "The Web of Silent Springs"

Sol. Q14. In "Thinking Like a Mountain", Leopold reflected on the death of a wolf, realizing the mountain depends on predators to prevent deer from over-browsing slopes and triggering severe soil erosion. **Answer: (B)**

Q15. Wendell Berry's agrarian ecological philosophy in **The Unsettling of America** centers primarily upon which core ethical duty?

(A) Replacing family farms with automated, satellite-guided robotic mega-combines
(B) Maximizing short-term export commodity yields through intensive chemical fertilizers
(C) Practicing intimate land stewardship, agrarian memory, and treating eating as an ecological act
(D) Encouraging complete rural depopulation into centralized high-tech urban megalopolises

Sol. Q15. Wendell Berry advocates for faithful land stewardship, localized family farming, deep affection for place (**topophilia**), and the recognition that "eating is an agricultural act." **Answer: (C)**